Initial Backtest Report

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Contents

1 History and Introduction of Strategy **2**

2 Description and Code **2**

2.1 Data 2

2.2 Strategy 3

2.3 Backtest 3

3 Performance of Strategy by Original Publisher **3**

4 Performance of Strategy in Backtest **4**

5 Backtesting Assumptions **5**

6 References **6**

7 Appendix **7**

1 History and Introduction of Strategy

The Average True Range (ATR) Rubber Band strategy is based on the Rubber Band trading strategy as described on the website Quantified Strategies [Grob]. The Rubber Band trading strategy is a contrarian principle and a mean-reversion strategy. It tries to buy when the market is oversold and sell when overbought, as it expects prices to revert to the mean. It can be utilised with many different indicators, and in this case we use the ATR. The ATR is a price volatility indicator which shows how much the price of the asset varies in a given time period on average. This strategy utilises the ATR to determine when the price is low and we should buy, and utilises the close and yesterday's high to determine when the price is high and we should sell. Examples of the backtests run on this strategy are on Quantified Strategies website [Groa] and medium blog by Quantified Strategies [Str].

2 Description and Code

The n-day ATR is calculated by taking $(High - Low)/n$.

The strategy is carried out as follows:

1. Calculate the n-day ATR.
2. Determine the High of the last n days.
3. Calculate a band which is 2.5 times below the n-day High using the n-day ATR.
4. If the close is below the band, we go long.
5. We exit from the position when the close is higher than yesterday's high.

In our backtest, we shall calculate the 5-day ATR. Hence we use $n = 5$ for the strategy steps above. The code for the backtest is attached below on page 7.

2.1 Data

We are using Python 3.12.5, pandas 2.2.2, yfinance 0.2.43, Backtesting.py 0.3.4, matplotlib 3.9.2.

We use the data from Yahoo Finance, and look at the asset SPDR S&P 500 ETF, (SPY). This is similar to the backtest performed by the original publisher. We define the date range for the backtest by considering that SPY was established 22 January 1993, and test it on data up to today. This is similar to the date range considered by the original publisher.

2.2 Strategy

We utilise the function "ATR" to calculate the Average True Range over a rolling window of n days. We also use the function "nHighs" to calculate the rolling n day High. We define the "ATRBand" Strategy class by setting the number of periods $n = 5$, and precomputing the values of the band and the previous day's High, thus completing steps 1-3 of the strategy.

We then define the iterator "next" by asking it to carry out steps 4 and 5 of the strategy. We first check if the close crosses below the ATR Band, and in that case buy to go long. Then we check if the close crosses above yesterday's High, in which case we sell off our position.

2.3 Backtest

We run the backtest of the ATRBand strategy on the data obtained on SPY. We assume that we start with a portfolio of $100,000, similar to what the original publisher did, and assume a commission rate of 0.2%.

3 Performance of Strategy by Original Publisher

The original publisher backtests the "SPY" ticker using $n = 5$ days as shown on the Quantified Strategies website on [Groa]. They backtest the strategy starting with $100,000 of equity, and the date starting in 1993, with the start date of the fund being 22 January 1993. We can see that they determined that the strategy has performed very well. At the end of the backtested period, they have a portfolio equity of $647625.2, the average annualised return at 6.4%, and the average gain per trade at 0.66%. The equity curve of their backtest is attached as shown in Figure 1 below. The original documentation of their test is in Reference [1]

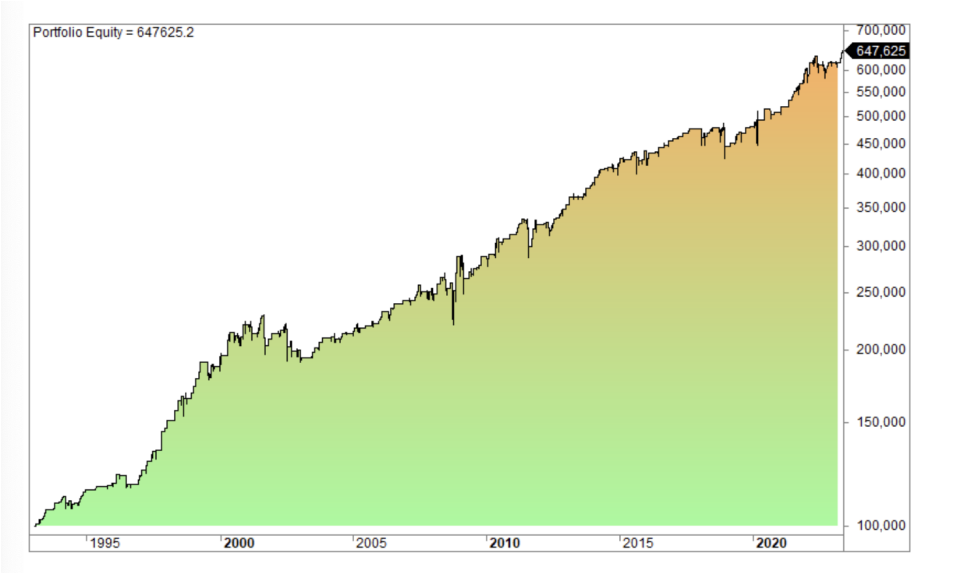

Figure 1: Plot of Equity Against Time for SPY Backtest

4 Performance of Strategy in Backtest

From our backtest, we can see that we obtain a fairly similar final portfolio equity and average return rate. We end up with $717122.8, and an average annualised return at 6.4% as well. The final portfolio equity in our backtest could be higher as we are likely running the backtest for a longer period of time. The exact dates of the backtest is not stated by the original publisher, and we have inferred from their equity curve Figure 1 on page 4 to obtain a similar range of dates. The average gain per trade is also similar, at 0.61% compared to their 0.66%.

We have also placed our equity graph, Figure 2 on page 5, on a log-based scale for the y-axis representing equity for an easier comparison. We can see that there are some differences on the equity line, particularly surrounding the large drawdown after 2008.

This could be due to differences in assumptions such as brokerage fees and position sizes when buying and selling made during the backtesting stage.

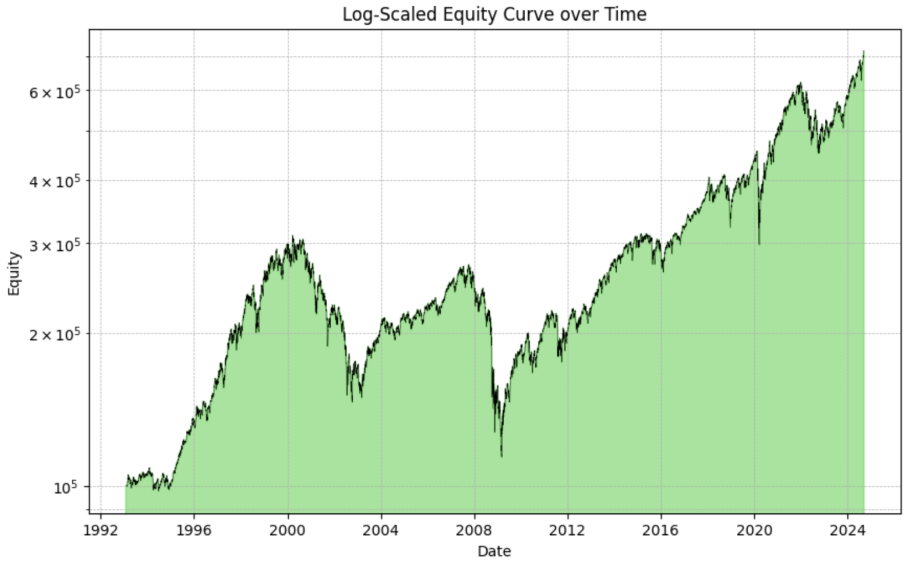

Figure 2: Plot of Equity Against Time for SPY Backtest

5 Backtesting Assumptions

In this backtest, we assume that the broker commission is 0.2% and we start with $100,000 of equity. We have downloaded data from 22 January 1993 but data from Yahoo Finance only starts from date 29 January 1993, and we end at the current data, 16 September 2024. Inherited from Strategy.buy() and Strategy.sell() in backtesting.py, we assume that the order size each time is 99.9% of the current portfolio.